Yun-Yen Wu

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Profile

Third-year physics undergraduate (GPA 4.20/4.30) with graduate-level Quantum Field Theory training. My research interests lie in **Quantum Field Theory**, **Conformal Field Theory** (CFT), and **non-Hermitian systems** in condensed matter contexts. Currently developing non-Hermitian systems in 1+1D under Prof. Chang-Tse Hsieh using **bosonization**.

Research Experience

- **Project: Non-Hermitian Bosonization and Logarithmic CFT** (presentation slide)
  2024 Jul. – present, supervised by Prof. Chang-Tse Hsieh, National Taiwan University.
  - Developed the **1+1D bosonization correspondence** for non-Hermitian fermionic field theories, identifying consistent operator mappings and spectral structures.
  - Constructed **Jordan-block operator representations** to explore their implications within LCFT.
  - Performed analytical calculations in Mathematica to verify field-theoretic predictions.
- **Project: Low-Temperature Electrical Measurements of Cobalt Disilicide and Niobium Superconductors**
  2022 Sept. – 2023 Jul., supervised by Prof. Juhn-Jong Lin, National Yang Ming Chiao Tung University.
  - Analyzed resistance-temperature data to identify the low temperature behavior of $CoSi_2$ and Niobium.

Awards and Honors

- **Silver Medal**, International Physics Olympiad (IPhO), 2022
- **Bronze Medal**, Asian Physics Olympiad (APhO), 2022
- **Dean's List**, National Taiwan University, 2024

Education

National Taiwan University, Physics undergraduate (GPA: 4.20/4.30) — 2023 Sept. – present

Courseworks

- **Quantum Field Theory I & II** *(A+, Graduate)*
  Final presentation: Applied 1+1D bosonization deriving fermion-boson correspondence and analyzed the sine-Gordon model.
- **Quantum Mechanics I** *(A+, Graduate)*
- **Relativity** *(A+, Undergraduate)*
  Final presentation: Derived null geodesic equations from a Hamiltonian approach to analyze wormhole gravitational lensing.

Skills

- **Programming:** LaTeX (Proficient), Mathematica (Symbolic calculation), Python (for data analysis and visualization)
- **Languages:** English (TOFEL 101, TOEIC 900); Mandarin (Native)